LESSON PLAN

MOCKUP SQUADRON ENVIRONMENT (MSE) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	21	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Measurement Of rotational speed using tacho generator and tacho probe.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Measurement Of rotational speed using tacho generator and tacho probe.
5. Specific Objectives	24.3.1.1 Trainee will be able to measure the rotational speed.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Tacho Generator and Tacho Probe installation and reading verification with safety precautions and instructor supervision.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a tachometer in aircraft systems?
(b) Define rotational speed in the context of engine components.
(c) List two common types of devices used to measure rotational speed.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Principles of Tacho Generator operation • Construction and types of Tacho Generators • Signal generation and frequency conversion • Connection points for Tacho Generator installation • Calibration and accuracy factors	L & Ds	PPT, IB	15 min
(b)	• Working principles of Tacho Probe technology • Installation procedures for Tacho Probes • Interpreting probe signals and output readings • Troubleshooting common probe failures • Safety precautions during probe handling	L & Ds	PPT, PDF	20 min
(c)	• Comparative analysis of Generator vs Probe methods • Error sources in rotational speed measurement • Maintenance requirements for measurement devices • Recording and logging measurement data • Integration with aircraft flight instruments	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference in signal output between a Tacho Generator and a Tacho Probe.
(b) Describe the steps required to safely install a Tacho Probe on an engine component.
(c) Identify three potential sources of error when measuring rotational speed.

12. Home Assignment and Follow Up	Prepare a brief report comparing the operational principles of Tacho Generators and Tacho Probes, referencing the content covered in LP 02 regarding basic engine instrumentation.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Demonstration and explanation on the operation of various elements of mechanics of an Altimeter, ASI, and VSI & Mach meter.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min